

AltScreen v5

Alt Alpha & Sensitivity Screener for TradingView Pine Screener - User Guide

September 2026. Companion to `alt_alpha_sensitivity_screener.pine` (v5) and `binance_perps_watchlist.py`.

1. What it does

AltScreen scans a watchlist of Binance USDT-M perpetuals and answers two separate questions about each name. They are answered by different columns and should be sorted separately; blending them into one score hides the information in both.

Question	What it isolates	Primary columns
Alpha - who is doing something BTC and its own cohort do <i>not</i> explain?	Return left over after removing (1) BTC beta and (2) the co-movement of a chosen cohort basket. Normalised as a t-statistic so high- and low-vol names are comparable.	Alpha t (2F), Emerging, Persist, Alpha on volume / thin
Sensitivity - who will overreact most on the next BTC impulse?	How much an alt's BTC beta expands on large BTC bars versus quiet ones (the second derivative of beta), plus how compressed and how long-quiet the alt currently is.	Amplification, Beta slope, Coil

Sensitivity columns are **not** alpha. They rank vehicles for a BTC directional view - which alt gives the most movement per unit of BTC move right now. Alpha columns rank names with idiosyncratic flow. A name that is high on both lists is the rare one worth sending to the tape tools.

2. Setup

2.1 Requirements

- TradingView **Premium** (Pine Screener is gated to Premium and above).
- A watchlist containing only `BINANCE:XXXUSDT.P` symbols. Mixed watchlists (stocks, futures) produce meaningless betas for the non-crypto rows.
- The v4 script saved in your Pine Editor and added to Favorites.

2.2 Install the script

- Open any chart, then the **Pine Editor** tab at the bottom.
- Create a new indicator, delete the template, paste the full contents of `alt_alpha_sensitivity_screener.pine`.
- Click **Save**. Confirm the console line reports no compile errors - the screener runs the last version that compiled, so a silent compile error looks like an empty table.
- Open **Indicators** on the chart, find AltScreen under *My scripts*, and click the star to add it to Favorites. The Pine Screener only lists favorited scripts.

2.3 Build the watchlist

Run the helper script on your machine (standard library only - nothing to install):

```
python binance_perps_watchlist.py --top 120
```

It queries TradingView's own symbol scanner for every BINANCE symbol ending in USDT.P, sorts by 24h traded value, and writes `binance_usdt_perps.txt` in TradingView's import format. Then in TradingView: watchlist panel, the ... menu at the top, **Import list...**, pick the file.

Use `--top 120` or `--min-usd-vol 30e6` for the tradeable set. Illiquid names produce spurious alpha: their residuals are microstructure noise and the t-stats will lie.

2.4 Open the Pine Screener

- Direct URL: `tradingview.com/pine-screener/`, or from the home page **Products - Screeners - Pine**. From inside any other screener, the type switcher at the top-left of the page lists **Pine**.
- **Source** pill: choose your Binance perps watchlist.
- **Indicator**: choose AltScreen from Favorites. The first ten plots populate as columns automatically; the remaining columns are toggled on with the column button at the top-right of the table.
- Set the **timeframe** (see section 5), then **Scan**. Re-scan after changing source, inputs, or timeframe.

3. Settings

Open the gear icon next to the timeframe pill in the screener toolbar (it opens the indicator's inputs). Numeric and dropdown inputs are honored by the screener. **Symbol inputs are not** - Pine Screener ignores `input.symbol()` and uses script defaults, which is why every symbol in v4 is a constant in the code.

Input	Default	Meaning
Regression window	144	Bars used for every rolling beta and loading estimate. Longer = more stable, slower to react. Must leave room inside the screener's 500-bar limit.
Alpha lookback	72	Bars over which the alpha t-stats, Persist, Emerging, and the volume/thin sums are computed. Roughly half the regression window works well.
Beta slope lookback	24	Bars back for the Beta slope column (beta now minus beta then).
Impulse threshold	1.5	A BTC bar counts as an impulse when <code> BTC return </code> exceeds this multiple of BTC's long realized vol. Lower = more impulse bars, noisier split; higher = fewer, cleaner.
Realized vol short / long	24 / 144	Windows for the Vol ratio and Coil columns and for the impulse classifier.
Cohort basket	Auto	Auto assigns each scanned name a sector basket from a map in the code, so one scan measures every name against its own group. Any other choice forces that basket for all names. See 3.1.

3.1 Cohort selection

The Pine Screener allows at most five `request.security()` calls per script. One goes to BTC; the other four are the cohort legs. A fully adaptive cohort (finding each name's real co-movers from a large universe by correlation) needs ~30 requests and cannot run in the screener. v5 does the next best thing: in **Auto** mode a sector map inside the script (about 90 names, keyed on the Binance ticker) assigns each scanned symbol to one of six baskets, and the four legs of that basket are requested per symbol. One scan therefore measures every name against its own group. Unmapped names default to Majors. Forcing a single basket is still available for "who is beating the majors" style scans.

The **Sector** column (1 Legacy L1, 2 Majors, 3 L2/New L1, 4 DeFi, 5 Memes, 6 AI) shows which basket a name was measured against. If a name is mapped wrongly for your purposes, edit its line in `f_sector()`. Names missing from the map show 2 (Majors).

Preset	Legs	Use when asking
Legacy L1	DOT, HBAR, ICP, INJ	Is this name outrunning the 2021-era L1/infra rotation it usually rides with? (AVAX's natural cohort.)
Majors	ETH, SOL, XRP, BNB	Is this name outrunning liquid large-cap alt beta?
L2 / New L1	ARB, OP, SUI, APT	Is this outrunning the newer-chain trade?
DeFi	AAVE, UNI, LINK, ENA	Is this outrunning the cash-flow DeFi cohort?
Memes	DOGE, 1000PEPE, WIF, 1000SHIB	Is this outrunning the attention/meme factor?
AI	TAO, FET, RENDER, WLD	Is this outrunning the AI narrative cohort?

A name that is itself a leg of its basket is automatically excluded from its own cohort (the basket runs on the other three legs, and **Cohort legs live** shows 3). To change a basket, edit the symbol strings in the `switch` block of the script; every leg must exist on Binance USDT-M.

4. Column reference

All returns are close-to-close on the screener timeframe, in basis points. "Window" is the regression window; "lookback" is the alpha lookback. Regression-based columns show "-" until a symbol has *window* bars of history, so newly listed names do not produce fake betas.

4.1 Alpha group

Column	What it computes	How to read it
Chg 24h %	Percent change in close over the number of bars that make up 24 hours on the screener timeframe (24 bars on 1h, 6 on 4h, 1 on 1D).	Plain price context. On 1W or 1M it degrades to a one-bar change because 24h is shorter than a bar.
Beta BTC	Rolling through-origin regression slope of the alt's returns on BTC's over the window.	1.0 = moves one-for-one with BTC in bps; 1.5 = 150% of BTC's move. This is the <i>average</i> beta and says nothing about impulse vs quiet.
Alpha t (vs BTC)	Mean of the residual after removing BTC beta only, divided by its standard error, over the lookback.	A t-statistic. t above ~2 is hard to get by chance; 1.5-2 is worth a look; under 1 is noise. Can be fooled by cohort rotation.
Cohort loading	After stripping BTC from both the alt and the cohort basket, the regression slope of the alt's residual on the cohort's residual.	Near 0: decoupled from this group. Near 1: trades as a member of it. Negative: trades against it.
Alpha t (2F)	Same t-stat after removing BTC <i>and</i> the cohort factor.	Primary alpha sort. High here and high vs BTC = name-specific. High vs BTC but low here = the cohort was rotating and this name was a passenger.
Persist (-3..3)	Lookback split into thirds. Sign = newest third's residual direction; magnitude = how many thirds agree.	+3 = bullish alpha in every third (established, possibly late). +1 = only the newest third (just flipped, could be noise). Negatives mirror for bearish alpha.
Emerging (+/-1)	Fires when the newest two thirds agree and the oldest disagreed.	The early-but-confirmed state: alpha turned and has held for two periods. Screen for Emerging = 1 with Alpha t (2F) > 1.5.
Alpha on volume bps	Sum of the 2F residual over the lookback, counting only bars with above-average volume.	Alpha that arrived with real participation.

Column	What it computes	How to read it
Alpha on thin bps	Same sum, bars with below-average volume.	Alpha that arrived with nobody trading - market makers repricing off beta. Alpha mostly on thin evaporates on the first real seller.

4.2 Sensitivity group

Column	What it computes	How to read it
Beta impulse	Beta estimated only on bars where BTC return exceeded the impulse threshold times BTC's long realized vol.	How much the alt does when BTC actually moves.
Beta quiet	Beta estimated only on the remaining bars.	How much it does in BTC chop. Usually lower and noisier; can sit near zero in dead ranges.
Amplification	Beta impulse divided by Beta quiet , with Beta quiet floored at 0.25 so the ratio stays readable.	The second derivative of beta. 1.0 = linear response; 2.0 = twice the beta on impulses as in chop. Sort descending for the names most primed to overreact.
Impulse bars in window	Count of BTC impulse bars in the window.	Confidence check on the two betas above. Below ~10 the impulse beta is estimated on too few bars to trust; above ~20 is usable.
Beta slope	Beta BTC now minus Beta BTC <i>slope</i> <i>lookback</i> bars ago.	Positive: sensitivity expanding now. Negative: contracting. Amplification says how convex a name is; slope says whether that convexity is switching on.
Vol ratio short/long	Alt's realized vol over the short window divided by its realized vol over the long window.	Under 1 = quieter than its norm. 0.5 = half normal vol. The compression half of the coil.
Bars since own impulse	Bars since the alt itself moved more than the impulse threshold times its own long realized vol.	The duration half of the coil. How long it has been sitting.
Coil	(1 - Vol ratio) x Bars since own impulse.	Quiet <i>and</i> long-quiet scores high; noisy or recently moved scores low or negative. A rank-friendly heuristic - sort by it, do not read the number as a quantity.

4.3 Diagnostics

Column	What it computes	How to read it
Sector	Numeric code of the basket the name was measured against: 1 Legacy L1, 2 Majors, 3 L2/New L1, 4 DeFi, 5 Memes, 6 AI.	In Auto mode, confirms the mapping. A 2 on a name you consider DeFi or a meme means it is unmapped; add it to f_sector().
Cohort legs live	Number of basket legs that returned data and are not the scanned symbol itself.	Should be 4, or 3 when the name is a leg. 0 means the basket symbols are not resolving - check the strings in the switch block.

Open interest is **not** a column in v4. It cost one of the five request slots, and it is better read in MMT against the Coil and Beta slope columns: Coil rising with OI rising is a spring being wound; Coil rising with OI falling is an asset nobody is positioned in.

5. Workflows

5.1 Timeframe

The screener calculates on the last 500 bars only, and the defaults (144 / 72 / 24) are sized for that. The timeframe sets what those bars mean:

Timeframe	Regression window covers	Alpha lookback covers	Job
1h	6 days	3 days	Early-alpha detection; intraday sensitivity. The default job.
4h	24 days	12 days	Swing-horizon alpha; slower, cleaner t-stats.
1D	~5 months	~2.5 months	Slow-money read. Consider window 90 / lookback 30 / slope 10 so more names have history.

5.2 Alpha sort

- Leave Cohort basket on Auto. Sort by **Alpha t (2F)** descending.
- Filter or eyeball **Emerging = 1** and **Persist** of 2 for the early, confirmed state; Persist 3 is established.
- Check **Alpha on volume** vs **Alpha on thin**. Prefer names whose alpha came on volume.
- Compare **Alpha t (vs BTC)** to **Alpha t (2F)**: a large drop between them means the group was moving, not the name.
- Optionally re-scan with a forced basket (e.g. Majors) to ask a different question of the whole list.

5.3 Sensitivity sort

- Sort by **Amplification** descending. Discard rows where **Impulse bars in window** is under ~10.
- Cross with **Beta slope** for names whose sensitivity is expanding now.
- Sort by **Coil** for the compressed, long-quiet set, then check OI in MMT for loading.
- This list ranks vehicles for a BTC view. It is not a list of names with alpha.

5.4 Where the other tools fit

The screener is the discovery layer; it sees price and volume across 100+ names but not the tape. The workflow that uses each tool for what it is good at:

Stage	Tool	Question
1. Shortlist	AltScreen (this script)	Which names have 2F alpha, or high amplification with a coil, right now?
2. Positioning	MMT	Is OI loading behind the coil? How crowded is the perp?
3. Cohort check	Chart-side companion (planned)	With 30 legs available on a chart, who does this name really co-move with, and does the alpha survive that decomposition?
4. Tape	aggr Flow / Participants templates	Is the residual on real aggressive flow or thin market-maker repricing? Who is doing it (size buckets, venues)?

6. Pine Screener limits the script is built around

From TradingView's help-center page "Pine Screener: key features and requirements". Each of these broke an earlier version of this script.

Limit	Consequence for AltScreen
Max five request.security() calls per script	BTC + four cohort legs, with the legs chosen per symbol from the sector map. No OI request, no correlation-based cohort, no broad market index in the screener.

Limit	Consequence for AltScreen
input.symbol() is ignored (defaults used)	All symbols are constants. Baskets are changed via the dropdown or by editing code, not via symbol pickers.
Only the last 500 bars are calculated	Defaults shrunk to 144 / 72 / 24. Very long windows leave few valid bars and blank many names.
First ten plots become columns automatically	The auto ten: Chg 24h, Beta BTC, Alpha t (vs BTC), Cohort loading, Alpha t (2F), Emerging, Alpha on volume, Alpha on thin, Amplification, Coil. Persist, the impulse/quiet betas, slope, vol ratio, Sector and legs live are enabled via the table's column button.
Supported timeframes only	1, 5, 15, 30 min; 1, 2, 4 h; 1D, 1W, 1M. No custom timeframes.
Only favorited indicators are selectable	Star the script in the Indicators dialog after saving.
One scan at a time across browser tabs	A scan running in another tab blocks this one.

7. Troubleshooting

Symptom	Likely cause	Fix
Every column blank, including Coil and Cohort legs live	Script did not compile (screener runs the last compiled version), or the script exceeds the five-request cap.	Check the Pine Editor console after saving. Count request.security calls; v4 has exactly five.
Cohort loading and Alpha t (2F) blank; Beta BTC populated	Cohort legs not resolving. Cohort legs live reads 0.	A basket symbol string is wrong or delisted. Fix the string in the switch block. If it persists with known-good symbols, the screener may be rejecting per-symbol (dynamic) request arguments: set Cohort basket to a fixed preset to confirm.
Cannot find where to change the cohort	Looking for a symbol picker. There is none - symbol inputs are not supported by the screener.	The gear icon beside the timeframe pill opens the inputs; Cohort basket is a dropdown there. In Auto mode nothing needs changing.
Most rows show "-" in regression columns	Names lack <i>window</i> bars of history on this timeframe (new listings, or a long window on 1D).	Expected for new listings. Otherwise shorten the window or move to a lower timeframe.
Chart looks like a tangle of lines when the script is added to a chart	Seventeen plots in different units on one scale. Expected.	The chart is not the surface for this script; use the screener table. A chart companion is a separate script.
A new listing shows an extreme or negative Beta BTC	Warm-up gate not reached (should show "-") or window is very short.	Keep the window at 100+ bars; treat any beta on under ~100 bars as unreliable.
Amplification looks absurd (10+)	Beta quiet near the 0.25 floor with few impulse bars.	Check Impulse bars in window; below ~10, ignore the row.
Alpha t high but both volume/thin sums tiny	Very low-vol name; the t-stat normalises away the size.	Real but small. Cross-check the bps sums before acting.

8. Method appendix

8.1 Returns

$r = (\text{close} / \text{close}[1] - 1) \times 10000$, in basis points, for the alt, BTC, and each cohort leg on the screener timeframe.

8.2 Through-origin beta

$\text{beta}(y \text{ on } x) = \text{SMA}(y \text{ x}, \text{window}) / \text{SMA}(x \text{ x}, \text{window})$. A regression slope with no intercept, which is the right form for return-on-return relationships where the mean return is ~ 0 over the window. The denominator is floored to avoid division by zero in dead BTC ranges.

8.3 Two-factor residual

```
cohRet = equal-weight mean of live, non-self cohort leg returns
res1 = altRet - beta(alt on BTC) x btcRet
cohRes = cohRet - beta(coh on BTC) x btcRet
gamma = beta(res1 on cohRes)
res2 = res1 - gamma x cohRes
```

The cohort is residualised against BTC first so that gamma measures loading on the cohort's BTC-independent component only; otherwise BTC would be counted twice.

8.4 t-statistic

$t = \text{mean}(\text{res}, \text{lookback}) / (\text{stdev}(\text{res}, \text{lookback}) / \sqrt{\text{lookback}})$. Vol-normalised so names of different volatility are comparable. Treat $|t|$ above ~ 2 as unlikely to be chance over the lookback; this is a heuristic, not a test with adjusted degrees of freedom.

8.5 Impulse split

A BTC bar is an impulse when $|\text{btcRet}| > \text{impulse threshold} \times \text{stdev}(\text{btcRet}, \text{rvLong})$. Beta impulse and Beta quiet are through-origin betas computed with the products summed only over their own subset of bars. Amplification is their ratio with $|\text{Beta quiet}|$ floored at 0.25, a judgment call to keep the ratio finite when quiet beta collapses.

8.6 Judgment calls to be aware of

- The 0.25 floor on $|\text{Beta quiet}|$ is arbitrary. It prevents blow-ups; it also caps Amplification for names whose quiet beta is legitimately tiny.
- Coil is a constructed heuristic, not a measured quantity. It is monotone in the two things it combines and nothing more.
- The sector map is a hand-maintained stand-in for a real correlation-based cohort, and it will drift as narratives change. The adaptive version requires the chart-side companion.
- Equal weighting inside a basket over-weights its higher-vol legs. The through-origin gamma absorbs the scale but not the mix.

This tool produces analysis, not recommendations. Nothing in it is investment advice.